

Railway Deployment Guide for AI Agent (Step-by-Step)

Goal

Deploy a simple AI agent (FastAPI / Python) from GitHub to Railway and obtain a live URL.

Prerequisites

- 1 GitHub repository with your code
- 2 Working FastAPI agent locally
- 3 requirements.txt file
- 4 .env file (not pushed to Git)

Step 1: Prepare Project Structure

```
project/  
■■■ main.py  
■■■ requirements.txt  
■■■ .env (not committed)  
■■■ Procfile
```

Example main.py:

```
from fastapi import FastAPI  
  
app = FastAPI()  
  
@app.get("/")  
def home():  
    return {"message": "Agent is running"}
```

Procfile:

```
web: uvicorn main:app --host 0.0.0.0 --port $PORT
```

Step 2: Push Code to GitHub

```
git init  
git add .  
git commit -m "initial commit"  
git branch -M main  
git remote add origin <repo-url>  
git push -u origin main
```

Step 3: Create Railway Project

Go to railway.app → Login → New Project → Deploy from GitHub Repo

Step 4: Add Environment Variables

```
OPENAI_API_KEY=your_key  
OPENAI_MODEL=gpt-4o-mini
```

Step 5: Redeploy

Click Deploy → Redeploy after adding variables.

Step 6: Generate Public URL

Go to Settings → Domains → Generate Domain.

Step 7: Test Application

Visit '/' or '/docs' endpoint to verify agent is running.

Common Errors & Fixes

- 1 App not starting → Check logs in Railway
- 2 Port error → Ensure Procfile uses \$PORT
- 3 Module not found → Update requirements.txt
- 4 Env vars missing → Add in Railway dashboard
- 5 Works locally but not on Railway → Check dependencies and entry file

Student Checklist

- 1 Code pushed to GitHub
- 2 Railway project created
- 3 Environment variables added
- 4 Deployment successful
- 5 Public URL working
- 6 /docs accessible
- 7 Agent responding